Decoding Visual Stimulus Identity from Population Activity in Early Visual Cortex

Machine Learning in Neuroscience - Final Project Report

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GitHub link to project: https://github.com/Yaelts2/vsdi-face-decoding/tree/main/figures/110209a%20late

AI Statement

Claude (Anthropic) was used for: (1) drafting and editing report text based on our results and analysis; (2) debugging Python code for the decoding pipeline and plotting scripts; (3) generating boilerplate cross-validation and permutation testing code structures, which were subsequently modified and validated by the team.

ChatGPT (OpenAI) was used for literature review assistance and explaining statistical concepts. All scientific decisions. Experimental design, analysis choices, interpretation of results were made by the team. All code was reviewed, tested, and understood by team members before inclusion.

Abstract

Early visual cortices (V1, V2) are typically associated with low-level feature processing, yet recent evidence suggests they may carry higher-order information about stimulus category. We analyzed voltage-sensitive dye imaging (VSDI) recordings from V1, V2, and V4 of macaque monkeys performing a face versus scrambled-image discrimination task. A linear Support Vector Machine decoder was trained on spatiotemporal population activity patterns using nested cross-validation with trial-level grouping. The model achieved up to 97% trial-level accuracy (AUC = 0.93) during the early feedforward response window (50–130 ms post-stimulus). Sliding-window analysis revealed two temporal phases: an early peak shortly after stimulus onset, followed by a later resurgence around 200 ms reflecting potential top-down feedback. Weight maps showed V1-clustered face-preferring pixels and V2/V4-distributed scrambled-preferring pixels, with systematic spatial reorganization over time.

(a) Learning Problem Description

Visual processing is traditionally conceptualized as hierarchical: primary visual areas V1 and V2 extract low-level features such as orientation, spatial frequency, and contrast, while higher-order areas progressively assemble these into object-level representations (Hubel & Wiesel, 1962; Felleman & Van Essen, 1991). Under this view, early visual cortex should carry minimal information about stimulus category. However, Ayzenshtat et al. (2012) challenged this using voltage-sensitive dye imaging (VSDI) in behaving monkeys, showing that V1 population responses during a face versus scrambled discrimination task contain two temporal phases: an early phase encoding local stimulus features and a later phase correlating with perceptual report and behavioral outcome. This suggests category-level information may already exist in early visual cortex.

Our goal was to decode stimulus identity (face vs. scrambled) from single-trial VSDI population activity using machine learning, and to characterize *when* and *where* category information emerges across V1, V2, and V4. This is motivated by the observation that condition-averaged activation maps for face and non-face stimuli look nearly identical (Figure 1), yet the underlying single-trial population patterns may be separable by a trained classifier.

Data

VSDI recordings were obtained from V1, V2, and V4 of macaque monkeys performing a two-alternative forced-choice face vs. scrambled-image discrimination task. In each trial, a grayscale stimulus (126×126 px, 3.6°×3.6° visual angle) was briefly presented (80–300 ms) while the animal reported stimulus category for reward. Each trial consists of 256 frames at 100×100 pixel spatial resolution and 10 ms temporal resolution, with labels indicating stimulus category and behavioral outcome. The dataset spans 4 recording days with ~4 sessions/day and ~30 trials/condition/session. We validated the pipeline on a single-session subset (Dataset 1, ~28 trials/condition) before scaling to a multi-session dataset (Dataset 2) to confirm generalizability.

(b) Description of Previous Solutions

Ayzenshtat et al. (2012) analyzed VSDI population responses to natural images in V1 using condition-averaged maps and signal detection theory (d' measures). They identified two response phases but did not apply single-trial machine learning decoding. Their analysis demonstrated that population-level signals carry category-related information but did not apply single-trial machine learning decoding or characterize the spatial organization of discriminative features at the pixel level. Our work extends their findings by applying a supervised SVM decoding framework to extract trial-level category information, enabling quantitative accuracy assessment and systematic characterization of the spatial and temporal structure of discriminative population signals.

(c) Method Used

Preprocessing

Raw VSDI signals were normalized to background fluorescence (ΔF/F) and blank-condition subtracted to isolate stimulus-evoked responses. Single-trial responses were baseline-normalized per pixel: subtract trial-specific baseline mean, divide by across-trial baseline standard deviation. A manually defined ROI mask excluded peripheral noisy pixels. No dimensionality reduction was applied; each frame's feature vector contained all ROI pixels (~4,000–6,000 features depending on the recording session).

Model: Linear SVM

We chose a linear Support Vector Machine for three reasons: (1) *interpretability*- linear weights map directly back to cortical pixels, enabling spatial visualization of discriminative information; (2) *effectiveness in high-dimensional, low-sample settings*- SVMs handle $p >> n$ well with appropriate regularization; (3) *alignment with prior neural decoding literature*. Classification was performed at the frame level, where each time frame constitutes one sample represented by its spatial activation pattern across all ROI pixels.

Cross-Validation and Statistical Testing

To prevent information leakage (frames from the same trial share temporal autocorrelation), we used GroupKFold cross-validation with trial-level grouping, all frames from a given trial were assigned to the same fold. Model evaluation used nested 10-fold CV: an inner loop selected the regularization parameter C via grid search; an outer loop estimated generalization accuracy. The optimal C consistently converged to 0.0001 across all folds, indicating strong regularization. Trial-level accuracy was computed via majority vote across frames within each trial. Statistical significance was assessed by permutation testing with trial-level label shuffling, preserving the frame-grouping structure to prevent artificial inflation of sample size.

Fixed-Window Analysis

Based on the population activation profile, we defined two temporal windows: an *early window* (frames 32–44, ~50–130 ms post-stimulus) corresponding to the first feedforward activation peak, and a *late window* (frames 60–70, ~300–400 ms) to test whether category information persists beyond the initial feedforward sweep, potentially reflecting top-down feedback.

Sliding-Window Analysis

A 50 ms (5-frame) sliding window was shifted in 10 ms (1-frame) increments from 120 ms pre-stimulus to 900 ms post-stimulus. Since the fixed-window analysis consistently identified $C = 0.0001$, hyperparameter selection was not repeated. Each window was evaluated with 5-fold GroupKFold CV, yielding a time-resolved accuracy profile across the full trial. A control analysis with shuffled labels was run identically to establish the null distribution.

Weight Map Analysis

SVM weight vectors were extracted per outer CV fold, averaged across folds, and reshaped into 2D cortical geometry. Positive weights indicate face-preferring pixels; negative weights indicate scrambled-preferring pixels. For the sliding-window analysis, weight maps were computed independently per temporal window to track the spatial evolution of discriminative information over time. Additionally, mean activation time courses were computed separately for the top 20% positive-weight and bottom 20% negative-weight pixel groups to validate the category-relevance of the spatial patterns.

(d) Results

Fixed-Window Decoding

The fixed-window SVM decoder was evaluated on two independent datasets. Results are summarized in Table

Table 1. Fixed-window decoding performance across datasets and temporal windows.

Condition	Frame Acc.	Trial Acc.	AUC	Non-face	Face
DS1 (030209a), Early (0-80 ms)	85 ± 4%	88 ± 5%	0.93	91%	86%
DS2 (110209a), Early (0-80 ms)	86 ± 3%	97 ± 2%	0.93	100%	93%
DS2, (1102091), Late (100-180 ms)	83 ± 4%	79 ± 4%	0.90	82%	75%

Non-face and Face columns report per-class correct classification rates.

On Dataset 1 (single session, proof-of-concept), the early-window decoder achieved 88 ± 5% trial-level accuracy with AUC = 0.93 (Table 1). On Dataset 2 (110209a), performance improved to 97 ± 2% trial-level accuracy with AUC = 0.93 (Figure 2a), with several folds reaching 100% accuracy and frame-level accuracy consistently exceeding 70%. The performance difference likely reflects improved signal-to-noise ratio in Dataset 2, resulting in a more stable and separable cortical representation despite identical analytical procedures.

The late-window analysis was conducted exclusively on Dataset 2, as its stimulus duration of 100 ms allowed for a meaningful post-stimulus offset window (100–180 ms after stimulus onset). Dataset 1, with a stimulus duration of 1000 ms, does not have a comparable post-stimulus period within the trial window and was therefore excluded from this analysis. The late-window decoder yielded 79 ± 4% trial-level accuracy (AUC = 0.90), demonstrating that category information persists post-stimulus offset (Table 1). Notably, some individual folds showed 100% trial accuracy even in this late window, while others dropped to 50–67% (Figure 3a), suggesting that the late-phase representation may be more variable across trials than the early-phase signal. In all analyses, the optimal C converged to 0.0001. A permutation test with 100 label shuffles confirmed that the observed trial-level accuracy was highly significant (two-tailed $p < 0.001$), demonstrating that decoding performance cannot be explained by chance.

Sliding-Window Temporal Dynamics

This analysis was performed on Dataset 2 (110209a). Sliding-window significance was assessed using 1,000 trial-level label permutations while preserving the cross-validation structure. For each time window, the real accuracy was compared against the shuffled null distribution using a two-tailed non-parametric rank-based test ($\alpha = 0.001$). The time-resolved decoding profile revealed a characteristic multi-phasic pattern (Figure 5). During the pre-stimulus baseline, accuracy fluctuated near chance (~50%). Following stimulus onset, accuracy rose sharply to a first peak of ~93–95% trial-level accuracy during the early feedforward window (~50–130 ms), followed by a transient dip and a second peak of ~85–89% around 200 ms post-stimulus, potentially reflecting top-down feedback. Accuracy remained above chance (65–80%) for several hundred milliseconds before gradually declining. A shuffled-labels control remained at chance throughout the entire trial period, confirming that the observed temporal dynamics reflect genuine stimulus-related population signals rather than methodological artifacts. Multiple contiguous windows significantly exceeded the null distribution, confirming reliable time-resolved category information beyond chance.

Spatial Organization of Discriminative Weights

Weight map analysis revealed distinct spatial organization across cortical areas (Figure 2c and figure 3c). In the early window, the top 20% positive (face-preferring) weights were spatially clustered in V1, while the bottom 20% negative (scrambled-preferring) weights were distributed across V2/V4. Per-fold weight maps showed high cross-fold consistency (low standard deviation), indicating stable discriminative spatial patterns. The mean activation time courses for positive- vs. negative-weight pixel groups diverged selectively during face trials (positive-weight pixels showing higher activation), while converging during non-face trials (Figure 4) providing functional validation of the learned spatial weights.

Sliding-window weight maps showed systematic temporal evolution (Figure 6): early discriminative weights concentrated in retinotopically organized V1 regions, transitioning to broadly distributed patterns across V2/V4 during the later phase. This spatial reorganization coincided with the second accuracy peak, suggesting a shift from local feature-driven encoding to distributed, potentially object-level representation.

(e) Conclusion

A linear SVM decoder applied directly to high-dimensional VSDI pixel space achieved up to 97% trial-level accuracy for face vs. scrambled classification, confirming that distributed population signals in early visual cortex carry robust category information invisible in condition-averaged maps. The two-phase temporal profile—an early feedforward peak (~93–95%) followed by a later resurgence (~85–89%)—aligns with the hypothesis that V1/V2 participate in both initial feature extraction and feedback-modulated category representation, consistent with Ayzenshtat et al. (2012). Weight map analysis provides complementary insight: V1-clustered weights during the early phase suggest the decoder leverages retinotopic feature differences, while the shift to distributed V2/V4 weights during the late phase reflects progression toward integrative, object-level processing.

Was the project successful? Yes. We decoded stimulus category with high accuracy, replicated the two-phase temporal structure across datasets, and demonstrated interpretable spatial weight dynamics across cortical areas. The pipeline generalized from a small proof-of-concept to a larger multi-session dataset with improved performance.

(f) References

Ayzenshtat, I., Gilad, A., Zurawel, G., & Slovin, H. (2012). Population response to natural images in the primary visual cortex encodes local stimulus attributes and perceptual processing. *J. Neurosci., 32*(40), 13971–13986.

Felleman, D. J. & Van Essen, D. C. (1991). Distributed hierarchical processing in the primate cerebral cortex. *Cereb. Cortex, 1*(1), 1–47.

Hubel, D. H. & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *J. Physiol., 160*(1), 106–154.

(g) Figures

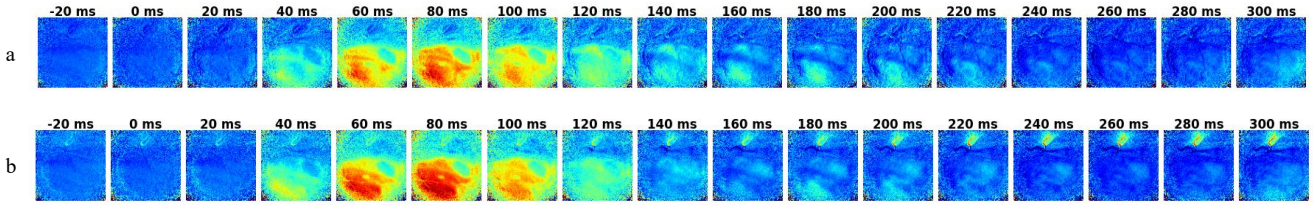

Figure 1. *Average VSDI activation maps for face (a - up) and scrambled (b- down) stimuli across time frames. Despite visually similar activation patterns, the underlying population signals are separable by machine learning.*

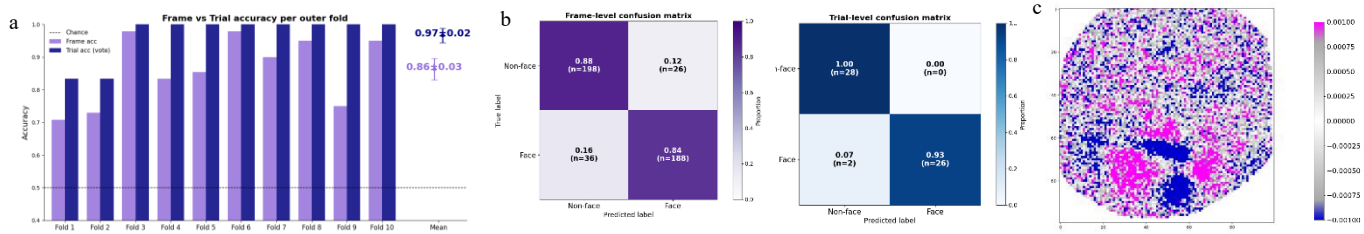

Figure 2. *Fixed-window decoding results (Dataset 2, early window). a: Frame- and trial-level accuracy per outer fold (mean trial accuracy = 97±2%). b : confusion matrix frame level and confusion matrix trial level. c :AVG weights map across folds (clip = (-0.001, 0.001)).*

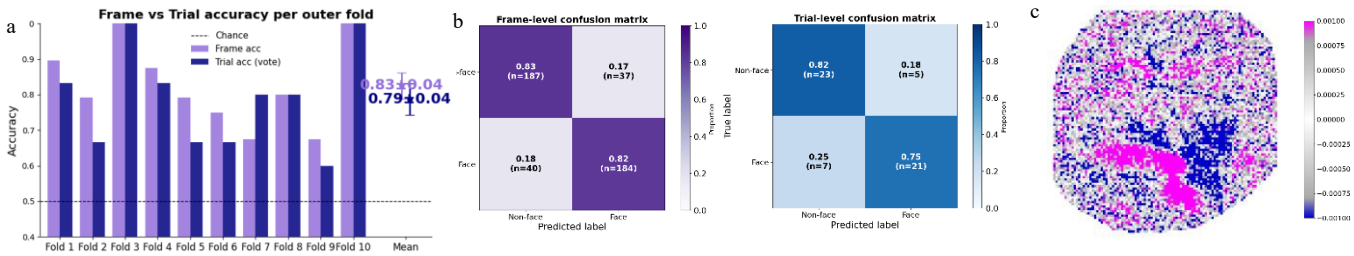

Figure 3. *Fixed-window decoding results (late window, 100-180 after stimuli onset). Accuracy per fold, a: Frame- and trial-level accuracy per outer fold (mean trial accuracy = 79±4%). b : confusion matrix frame level and confusion matrix trial level. c :AVG weights map across folds (clip = (-0.001, 0.001)).*

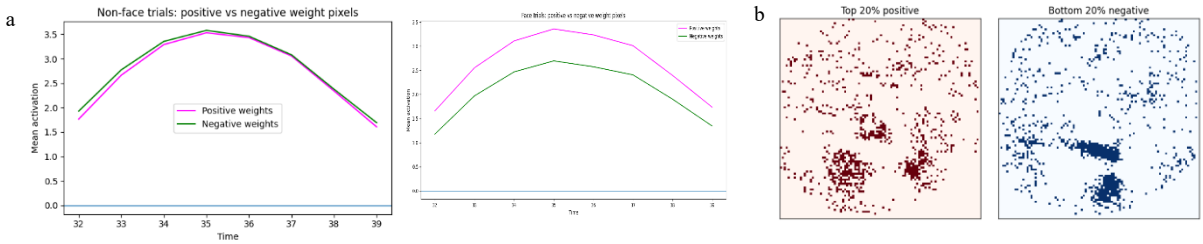

Figure 4. *a: show the average activation across the early window (0-80 ms) for pixels belonging to the model's positive and negative weight groups. b: display the spatial distribution of the top 20% positive weights and bottom 20% negative weights, highlighting the cortical regions that contribute most strongly to the decoder.*

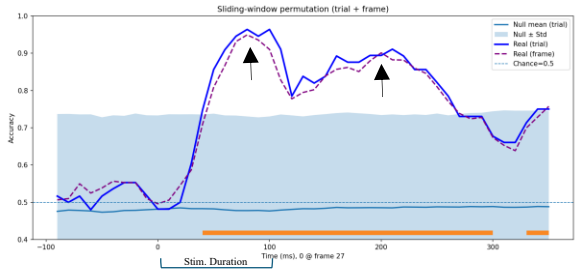

Figure 5. *Sliding-window decoding accuracy over time. Real data (trial-level) exhibits two prominent temporal peaks (early ~95%, late ~85–89%), whereas the shuffled-label control (mean of 1,000 permutations) remains at chance throughout. Orange bars indicate time windows that reached statistical significance (two-tailed permutation-based test, α = 0.001).*

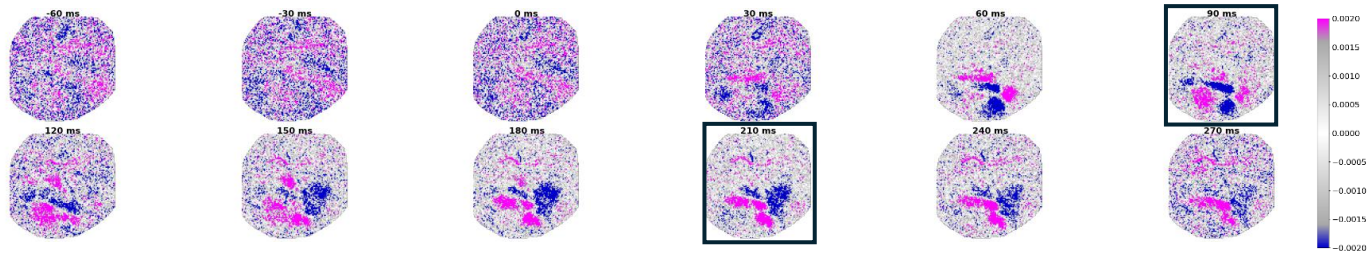

Figure 6. *Spatial weight maps are shown from −60 to 270 ms relative to stimulus onset. Structured positive and negative weight regions emerge after stimulus onset and intensify during the response period. Maps highlighted in squares correspond to time points of peak decoding accuracy, indicating that maximal classification performance coincides with strongly organized, spatially coherent informative patterns.*